

# DSA0602 • DATA HANDLING AND VISUALIZATION

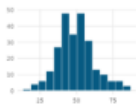
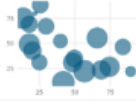
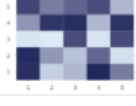
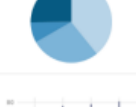
## VISUALIZATION MATCHING GAME - STUDENT HANDOUT

SESSION — ACTIVITY (15-20 MIN) • STUDENT HANDOUT • SLOT C

### Visualization Matching Game

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Each chart on the left (A-J) matches exactly one description on the right (1-10). Write the matching number next to each letter's blank.

|    |                                                                                     |                                                                          |
|----|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| A. |    | 1. _____<br>Compares distributions across groups and highlights outliers |
| B. |    | 2. _____<br>Shows how parts contribute to a whole                        |
| C. |    | 3. _____<br>Compares amounts across separate categories                  |
| D. |    | 4. _____<br>Shows magnitude across two categories using color intensity  |
| E. |    | 5. _____<br>Shows three variables at once using position and size        |
| F. |   | 6. _____<br>Shows how a value changes over time                          |
| G. |  | 7. _____<br>Shows part-to-whole composition across categories            |
| H. |  | 8. _____<br>Shows the relationship between two continuous variables      |
| I. |  | 9. _____<br>Shows the distribution of a single continuous variable       |
| J. |  | 10. _____<br>Shows how a cumulative total changes over time              |

DSA0602 • DATA HANDLING AND VISUALIZATION  
ASSESSMENT RECORD

**Visualization Matching Game | Slot C | Session Activity**

|                           |                                               |
|---------------------------|-----------------------------------------------|
| Department                | AIML                                          |
| Course Code / Course Name | DSA0601 / Data Handling and Visualization     |
| Experiment                | Visualization Matching Game (Slot C Activity) |
| Name                      | <b>BANDLAPALLI BHANUTEJA REDDY</b>            |
| Register Number           | <b>192325016</b>                              |
| Date                      | 03 OCTOBER 2026                               |
| Day                       | Thursday                                      |

**Aim**

To match each of the ten charts (A to J) in the handout with the one description that fits it best, and to explain in my own words why each match is correct.

**Marks Distribution (100 Marks)**

| Part     | Content                                             | Marks |
|----------|-----------------------------------------------------|-------|
| <b>A</b> | Final matching answer key (10 matches x 2 marks)    | 20    |
| <b>B</b> | Explanation of each match (10 charts x 6 marks)     | 60    |
| <b>C</b> | Comparison of look-alike charts (4 pairs x 5 marks) | 20    |
|          | Total                                               | 100   |

**PART A:**

**Question: Write the matching number next to each chart letter.**

| Chart    | Chart Type   | Match No. | Matching Description                                   |
|----------|--------------|-----------|--------------------------------------------------------|
| <b>A</b> | Histogram    | 9         | Shows the distribution of a single continuous variable |
| <b>B</b> | Line chart   | 6         | Shows how a value changes over time                    |
| <b>C</b> | Bubble chart | 5         | Shows three variables at once using position and size  |

|          |                         |    |                                                              |
|----------|-------------------------|----|--------------------------------------------------------------|
| <b>D</b> | Heatmap                 | 4  | Shows magnitude across two categories using color intensity  |
| <b>E</b> | Stacked bar chart       | 7  | Shows part-to-whole composition across categories            |
| <b>F</b> | Bar chart               | 3  | Compares amounts across separate categories                  |
| <b>G</b> | Area chart (cumulative) | 10 | Shows how a cumulative total changes over time               |
| <b>H</b> | Scatter plot            | 8  | Shows the relationship between two continuous variables      |
| <b>I</b> | Pie chart               | 2  | Shows how parts contribute to a whole                        |
| <b>J</b> | Box plot                | 1  | Compares distributions across groups and highlights outliers |

**Result:** All ten charts are matched, and no description is used twice. The final order is A-9, B-6, C-5, D-4, E-7, F-3, G-10, H-8, I-2, J-1.

### **PART B: Explanation of Each Match**

Each chart is explained by what I could see in the picture, the description it matches, and a simple real-life use.

#### **Chart A: Histogram**

**Question:** Which description matches chart A?

**Answer:** 9. Shows the distribution of a single continuous variable

**Explanation:** The bars in chart A touch each other and rise into a bell-like hump in the middle. Each bar stands for a range of values (a bin) of one numeric variable, and its height tells how many observations fall in that range. That is exactly what a distribution looks like, so it matches description 9.

**Real-life use:** Spread of student marks or customer ages.

**Chart B: Line chart**

**Question:** Which description matches chart B?

**Answer:** 6. Shows how a value changes over time

**Explanation:** Chart B has one line joining points that move up and down from left to right, and the x-axis runs in steps like 1, 3, 5, 8. When the x-axis is time, the line lets us follow the rise and fall from one period to the next, which is why it matches description 6.

**Real-life use:** Monthly sales or daily temperature.

**Chart C: Bubble chart**

**Question:** Which description matches chart C?

**Answer:** 5. Shows three variables at once using position and size

**Explanation:** In chart C every point is a circle of a different size. The left-right position gives one variable, the up-down position gives a second one, and the size of the circle gives a third. Showing three variables in a single chart is the key clue for description 5.

**Real-life use:** Advertising spend (x), sales (y) and profit (bubble size).

**Chart D: Heatmap**

**Question:** Which description matches chart D?

**Answer:** 4. Shows magnitude across two categories using color intensity

**Explanation:** Chart D is a grid of small squares in light and dark shades. The rows are one category, the columns are another, and the darkness of each square tells how big the value is. Using color intensity to show magnitude across two categories matches description 4.

**Real-life use:** Number of visitors by day of the week and hour of the day.

**Chart E: Stacked bar chart**

**Question: Which description matches chart E?**

**Answer:** 7. Shows part-to-whole composition across categories

**Explanation:** Each bar in chart E is split into two shades piled on top of each other, and there is one bar for each of the categories C1 to C5. The full bar height gives the total, and the two shades show how that total is made up. This is part-to-whole composition across categories, so it matches description 7.

**Real-life use:** Sales of two products in each region.

**Chart F: Bar chart**

**Question: Which description matches chart F?**

**Answer:** 3. Compares amounts across separate categories

**Explanation:** Chart F has separate bars with gaps between them, one for each of C1 to C6, and the height of a bar shows the amount for that category. Since the bars are for different, independent categories and are meant to be compared, it matches description 3.

**Real-life use:** Units sold for each product.

**Chart G: Area chart**

**Question: Which description matches chart G?**

**Answer:** 10. Shows how a cumulative total changes over time

**Explanation:** Chart G is a filled area under a line that keeps climbing and never goes down. A line that only rises is the mark of a running (cumulative) total, and the shaded area makes the growing amount easy to feel. That is why it matches description 10 and not a normal line chart.

**Real-life use:** Total downloads or total sales added up month after month.

**Chart H: Scatter plot**

**Question: Which description matches chart H?**

**Answer:** 8. Shows the relationship between two continuous variables

**Explanation:** Chart H is a cloud of small dots, and the dots drift upward from left to right. Each dot is one observation with a value on the x-axis and a value on the y-axis, and the upward drift suggests a positive relationship. So it matches description 8.

**Real-life use:** Study hours against exam marks.

**Chart I: Pie chart****Question: Which description matches chart I?****Answer:** 2. Shows how parts contribute to a whole**Explanation:** Chart I is a circle cut into slices of different sizes in different shades. The whole circle is 100 percent and each slice is one part's share of it, so it matches description 2.**Real-life use:** Share of the market held by different companies.**Chart J: Box plot****Question: Which description matches chart J?****Answer:** 1. Compares distributions across groups and highlights outliers**Explanation:** Chart J has a box with a middle line and thin whiskers for each group S1 to S4. The box shows the middle spread of the data, the line shows the median, and the whiskers (and any points beyond them) show the range and possible outliers. Because it does this for several groups side by side, it matches description 1. In the picture, group S1 clearly has a lower median than the others.**Real-life use:** Marks of different sections of a class.**PART C: Comparing Look-alike Charts (20 Marks)**

Some charts look similar, so I compared four pairs to be sure I had not mixed them up.

**1. Histogram (A) and Bar chart (F) (5 Marks)****Result:** Both use bars, but the histogram bars touch because the x-axis is a continuous number line split into ranges, while bar chart bars have gaps because they stand for separate categories. So A shows a distribution (9) and F compares categories (3).**2. Line chart (B) and Area chart (G) (5 Marks)****Result:** The line chart shows the value of each period, which can go up or down. The area chart in this handout only climbs and is filled, which shows a cumulative total. So B is 6 and G is 10.**3. Scatter plot (H) and Bubble chart (C) (5 Marks)****Result:** Both place points using two variables, but the bubble chart adds a third variable through the size of the circle. So H is 8 and C is 5.**4. Pie chart (I) and Stacked bar chart (E) (5 Marks)****Result:** A pie chart shows the parts of one single whole. A stacked bar chart shows the parts of the whole for several categories at once, so the totals can also be compared. So I is 2 and E is 7.

**Conclusion**

Every chart type answers a different kind of question: distributions, trends over time, relationships, comparisons between categories, and parts of a whole. Choosing the chart by the question we want to answer, and not by how it looks, is the main idea of this activity.